Design and Implementation of a Non-coupled Integrated Transformer for a Two-phase Interleaved Flyback PFC Rectifier

Karl Raymond Roque, *Student Member, IEEE*, Haruki Hirasawa, Sihoon Choi, *Member, IEEE*, Yu Yonezawa, *Member, IEEE*, Jun Imaoka, *Member, IEEE*, and Masayoshi Yamamoto, *Member, IEEE*

***Abstract*—Universal serial bus power delivery (USB PD) is a widely-applied standard for electronic device charging. In 2021, USB PD 3.1 was announced with a variable output of 5 V to 48 V and a maximum output power of 240 W. A multi-phase flyback rectifier is capable of power factor correction, galvanic isolation, and wide output voltage range, and is proposed as an alternative topology for USB PD. However, the number of magnetic components increases with the number of phases, which can lead to lower power density. This paper proposes the design of a non-coupled integrated transformer for a two-phase interleaved flyback PFC rectifier to reduce the size and number of magnetic components. Independent operation of the two phases using the proposed transformer was verified through simulation and experiments using a 240-W prototype operating at 100 kHz in discontinuous conduction mode (DCM). After conducting a volume analysis through the area product (AP) method, the proposed transformer is estimated to achieve up to 56% volume reduction compared to its conventional counterpart.**

***Index Terms*—Integrated magnetics, Transformer design, Two-phase interleaved flyback rectifier, USB PD, PFC, Core volume analysis.**

I. Introduction

UNIVERSAL serial bus power delivery (USB PD) is a technology used for convenient, portable, fast charging in electronic devices. The USB PD 3.1 standard, announced in 2021, expanded the maximum supplied power up to 240 W, allowing for higher power applications [1]. Adapters for USB PD 3.1 must be capable of power factor correction (PFC) for supporting 75 W and above, be equipped with galvanic isolation for user safety, and be capable of providing an output voltage of 5 V to 48 V [2]. Furthermore, USB PD 3.1 incorporates an adjustable voltage supply (AVS) to optimize charging efficiency within the extended power range (EPR), which is designated for power delivery exceeding 100 W up to a maximum of 240 W. The recent USB PD 3.2 standard, announced in 2023, further emphasizes this capability by expanding AVS mandates to the standard power range (SPR), which encompasses applications operating at or below 100 W [3].

A flyback rectifier has been widely used for commercial adapters due to its simple configuration, easy PFC control, and wide output voltage range. Improved configurations of flyback rectifiers such as active clamp flyback converters [4], [5] and bridgeless flyback rectifiers have also been proposed for higher converter efficiency [6], [7], [8], [9]. However, they are generally more practical for lower power applications due to the inherent high current stress on the switches by these topologies.

Several power supply configurations for USB PD have been proposed to meet the requirements. One example is a combination of a PFC rectifier, an LLC converter, and a buck converter as shown in Fig. 1(a) [10], [11]. While this configuration achieves high power density, the additional buck stage required to compensate for the limited gain range of the LLC increases the total component count and system cost. An asymmetrical half bridge (AHB) flyback converter in the second stage of a boost PFC circuit was introduced as shown in Fig. 1(b) [12]. The AHB allows for zero-voltage switching (ZVS) by the application of resonant capacitors in series placed on the primary side of the transformer. As a single-phase topology, the AHB processes the entire load power through a single transformer and switch pair. While this minimizes the component count, it inherently concentrates thermal stress in specific circuit elements.

Interleaving rectifiers have been considered to increase the maximum output power. Two-phase interleaving allows half of the total power to be supplied in each phase, therefore reducing the possible stress on the circuit elements [13]. The topology is shown in Fig. 1(c), where the two phases are parallel in the input and output side. The source nodes of switches S_1 and S_2 are common, and only one power supply is required for gate drive circuits. A common cathode package can be used for diodes D_1 and D_2, therefore fewer components are used for the circuit.

Operating multi-phase flyback converters across a wide output range presents some challenges. For instance, to ensure high efficiency across wide-range applications, converters must commonly employ phase-shedding, for which various

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K. Roque is with the Department of Electrical Engineering, Graduate School of Engineering, Nagoya University, Nagoya 464-8601, Japan (e-mail: roque.karl.raymond.dela.cruz.g4@s.mail.nagoya-u.ac.jp).

S. Choi, Y. Yonezawa, J. Imaoka and M. Yamamoto are with the Institute of Materials and Systems for Sustainability, Nagoya University, Nagoya 464-8601, Japan (e-mail: choi.sihoon.w4@f.mail.nagoya-u.ac.jp; yonezawa.yu.c9@f.mail.nagoya-u.ac.jp; imaoka.jun.n9@f.mail.nagoya-u.ac.jp; yamamoto.masayoshi.n5@f.mail.nagoya-u.ac.jp).

H. Hirasawa is with Toyota Industries Corporation, Toyota 448-8671, Japan (e-mail: haruki.hirasawa@mail.toyota-shokki.co.jp).

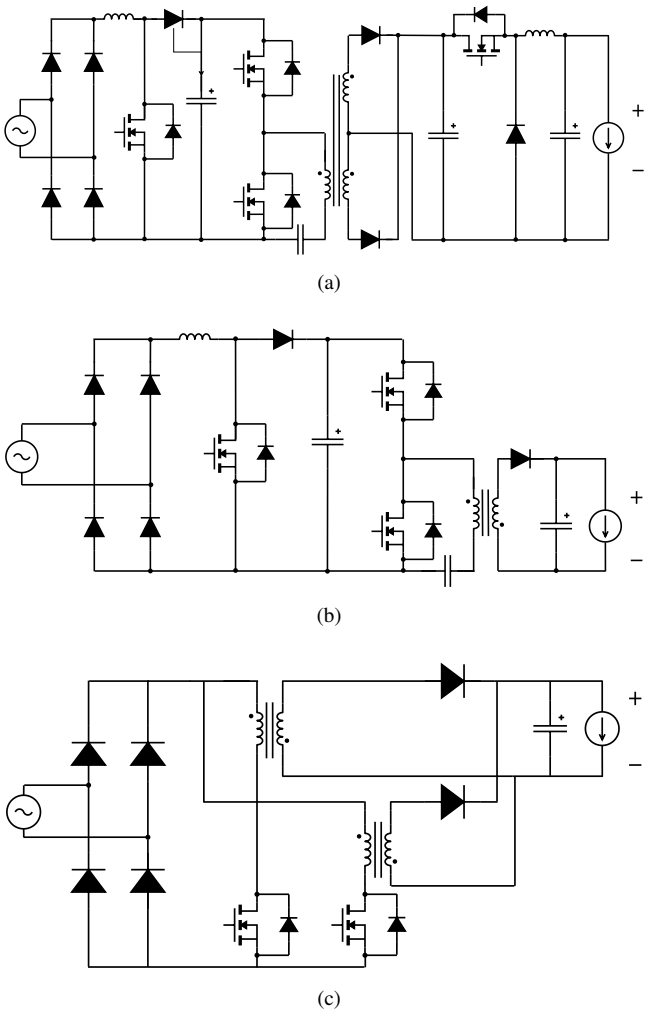


Fig. 1. Power supply configurations proposed for PD applications: (a) PFC boost, LLC, and buck converter combination in [10], [11], (b) PFC boost with AHB flyback [1], (c) two-phase interleaved flyback PFC rectifier.

control schemes have been proposed [14], [15], [16]. This strategy deactivates specific operational phases depending on the load condition to optimize light-load efficiency. However, the implementation of these control schemes often necessitates complicated designs and additional active components, thereby increasing the circuit complexity.

Moreover, increasing the number of phases inherently means an increase in the magnetic components required for the circuit. In the topology utilized for this study, although fewer active components are used overall, two discrete transformers are required, which compromises the power density of the converter. Integrated magnetics have been proposed and widely studied to improve the power density of various types of converters. Because integrated magnetics consolidate multiple discrete components through separate windings on a single magnetic core, these components inherently share the same flux paths. Consequently, the winding structure requires careful design to control magnetic flux. While magnetic coupling offers certain advantages, it introduces operational limitations. When mutual coupling exists between windings, the phases influence one another. In single-phase operation, when the phases are highly coupled, the active phase inevitably induces a voltage across the inactive phase, generating circulating currents that reduce light-load efficiency [17], [18]. Even under heavy-load conditions with multiple active phases, coupling can facilitate power delivery across non-operational phase intervals, leading to input current distortion and additional induced losses, therefore reducing overall efficiency [17].

For heavy-load applications, however, magnetic coupling can be beneficial. Coupling passively forces current equalization and minimizes capacitive current ripple, while the integrated structure improves power density. This intentional coupling is utilized in fixed-load server power supplies and LLC resonant converters [19], [20], [21]. Furthermore, modern interleaved flybacks require additional active switching devices for clamp circuits to ensure this current sharing [22]. Ultimately, while coupled magnetic structures are effective for fixed heavy-load targets, they conflict with the independent phase operation required by wide-range, variable-power converters. Other integrated structures are designed for alternative objectives, such as noise reduction via active cancellation within the windings [23].

Structural decoupling approaches have been explored to bypass these circulating currents and accommodate the variable-load requirements of USB PD, offering an alternative to control-based phase shedding. One approach proposes a four quadrants integrated transformer (FQIT) for a two-phase converter. This design decouples phases by separating the flux paths using a 3-D orthogonal-flux core geometry [24]. However, this custom core constrains commercial manufacturability and increases fabrication costs. Another proposed solution is an integrated nonlinear coupled inductor designed to decouple four phases at light loads [18]. Achieving this variable coupling requires customized, multi-piece L- and T-shaped core assemblies with manually configured geometries, introducing tight assembly tolerances and manufacturing complexity.

This paper proposes a non-coupled integrated transformer for a two-phase interleaved flyback rectifier as shown in Fig. 2(a) to address these limitations. The structure is designed to achieve zero magnetic coupling between phases, ensuring independent phase operation without magnetic interference. The proposed winding structure is shown in Fig. 2(b) where N_{ab} is the number of turns between terminals a and b. Half of N_{12} and N_{34} are wound at each outer leg with opposite polarity, so as to allow for flux cancellation at the center leg. On the other hand, N_{56} and N_{78} are also wound with opposite polarity on the center leg, and allows for the transformer N_{12} and N_{34} to not operate during its operation.

The zero-coupling coefficient between the two separate transformers simplifies implementation of rectifiers and transformer design. The operation of the rectifiers in each interleaved phase still follows that of a conventional flyback converter, preserving simplicity in control and timing. The design is also highly dependent on the duty cycle, an easily adjustable parameter in circuit design. The converter in this study aims to operate in DCM in order to allow for easier implementation of PFC and to conform to the standards of USB PD. It also realizes independent phase operation,

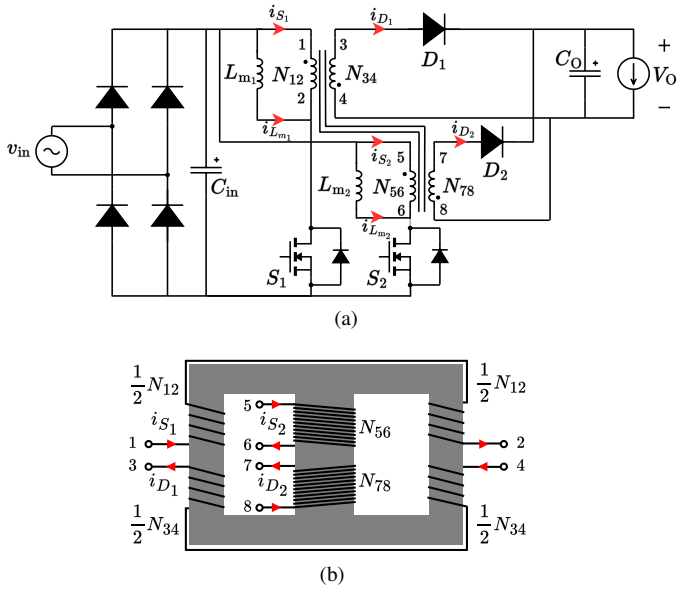


Fig. 2. The two-phase interleaved flyback rectifier (a), with the transformer and its winding structure in (b).

allowing the circuit to function in one-phase mode if desired, thereby enhancing flexibility in circuit operation and allowing simple implementation of phase-shedding. Furthermore, this entirely decoupled winding structure is easily achievable using a conventional EE-core, successfully bypassing the manufacturing complexities of custom geometries.

This article is an extended research of [25]. The preliminary study in [25] focused on the foundational design and experimental validation of the non-coupled integrated transformer. Key additions include a more detailed theoretical analysis, simulation validation of the operational principle, transformer design methodology including air-gap design, and a comprehensive core volume analysis utilizing the area product (AP) method. The rest of this paper is organized as follows: Section II presents an operational analysis of a two-phase interleaved flyback rectifier with the integrated transformer, along with its design method. Section III describes the size reduction effect of the transformer. Sections IV and V discuss the results of the simulation and experimental verification respectively, and Section VI presents the conclusions of this paper.

II. Design Method of the Proposed Transformer

A. Operation Analysis of the Proposed Flyback Rectifier and Transformer

The proposed integrated transformer, shown in Fig. 2(b) was designed for a two-phase interleaved flyback PFC rectifier, shown in Fig. 2(a) operating in discontinuous conduction mode (DCM). The switches (S_1, S_2 in Fig. 2(a)) operate with a fixed duty cycle D and a 180-degree phase shift.

Half of N_{12} and N_{34} are wound at each outer leg with opposite polarity. Moreover, N_{56} and N_{78} are also wound with opposite polarity at the center leg. The transformer utilized an EE core with a center leg that has a cross sectional area twice its outer leg, in accordance with conventional manufacturing specifications. The magnetizing inductances for both transformers $N_{12}-N_{34}$ and $N_{56}-N_{78}$ were set to be equal, and are denoted as L_{m_1} and L_{m_2} respectively. The saturation magnetic flux in each separate transformer during each phase is expected to be consistent.

The operational principle of the rectifier with the proposed transformer is illustrated in Fig. 3. Additionally, Fig. 4 presents the key waveforms of the six distinct operation modes. The rectifier is designed with duty cycle $D \leq 0.5\frac{nV_o}{V_{in_{max}}}$ with n as the turns ratio, V_o as the output voltage, and $V_{in_{max}}$ as the peak value of the input voltage. This condition ensures the magnetizing current of one phase returns to zero before the other phase peaks. This duration is defined as the discharge time t_{dis} and is given in terms of circuit parameters by (1):

$$t_{dis} = \frac{V_{in_{max}} D T_s}{n V_o} \tag{1}$$

with T_s as the switching period. For the condition given by D in this analysis, t_{dis} corresponds to the interval between t_3 and t_5 in Fig. 4.

To describe the magnetic state of the core, the superposition principle is used, as shown in Fig. 5(a). The magnetic flux density component B' is generated by the positive cycle transformer defined as $N_{12}-N_{34}$. The magnetic flux density component B'' represents the component generated by the negative cycle transformer defined as $N_{56}-N_{78}$. The total flux in any leg is the algebraic sum of these components. This superposition determines the peak flux density required for the core design, which is analyzed in detail in Section II-B.

Modes 1, 2, and 3 in Figs. 3(a)-(c) cover the operation during the positive AC input into the rectifier, while Modes 4, 5, and 6 in Figs. 3(d)-(f) describe the operation during the negative AC input. For the following modes, B'_{o_1}, B'_{o_2} and B'_c represent the magnetic flux densities in the respective left, right, and center legs of the core caused by the transformer active in positive AC input $N_{12}-N_{34}$, respectively, which are shown in Fig. 5(b). On the other hand, B''_{o_1}, B''_{o_2} and B''_c represent the magnetic flux densities in the respective left, right, and center legs of the core caused by the transformer active in negative AC input $N_{56}-N_{78}$, respectively, which are detailed visually in Fig. 5(c).

The detailed analysis of each operating mode is as follows:

1) Mode 1 ($t_1 < t < t_2$, Fig. 3(a)): At time t_1, switch S_1 turns on and S_2 remains off. The transformer $N_{12}-N_{34}$ begins storing energy in its magnetizing inductance. Its magnetizing current $i_{L_{m_1}}$ and the corresponding flux densities in the outer legs B'_{o_1} and B'_{o_2} linearly increase from zero. Simultaneously, transformer $N_{56}-N_{78}$ completes its discharge phase, with its magnetizing current $i_{L_{m_2}}$ and secondary diode current i_{D_2} decreasing to zero at t_2.
2) Mode 2 ($t_2 < t < t_3$, Fig. 3(b)): Mode 2 starts right after transformer $N_{56}-N_{78}$ completes its discharging phase, which means its associated currents i_{D_2}, $i_{L_{m_2}}$, and flux densities B''_{o_1}, B''_{o_2}, and B''_c are zero. Now, with S_1 still on, transformer $N_{12}-N_{34}$ continues storing energy until

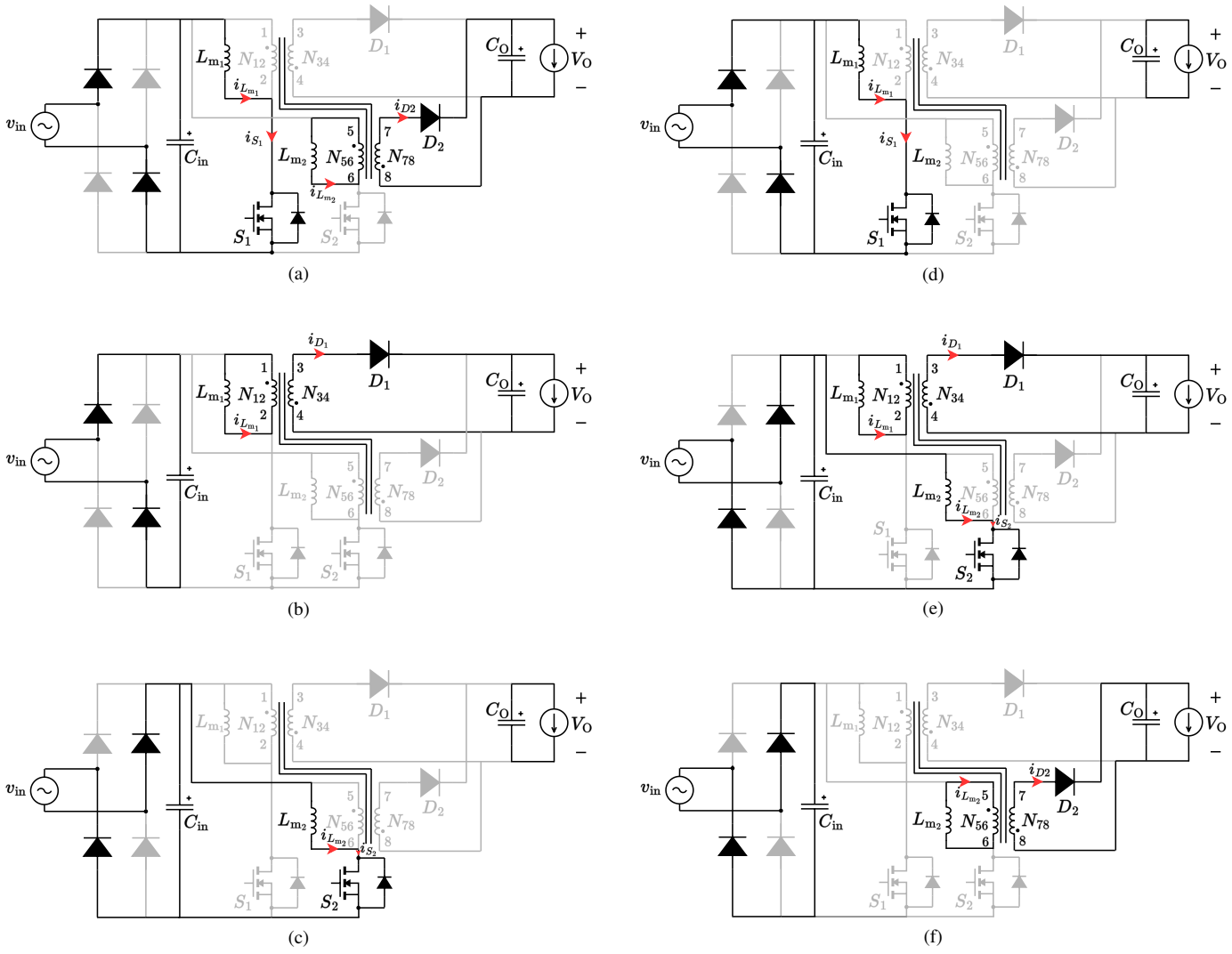


Fig. 3. Operational principle of the two-phased interleaved flyback rectifier with the proposed integrated transformer at $D \leq 0.5\frac{nV_o}{V_{in_{max}}}$. There are six distinct operational modes shown: (a) Mode 1 ($t_1 < t < t_2$), (b) Mode 2 ($t_2 < t < t_3$), (c) Mode 3 ($t_3 < t < t_4$), (d) Mode 4 ($t_4 < t < t_5$), (e) Mode 5 ($t_5 < t < t_6$), and (f) Mode 6 ($t_6 < t < t_7$).

$i_{L_{m_1}}$ reaches its peak value at t_3. The peak primary current in the positive cycle $i_{S_{1_{pk}}}$ is given by (2):

$$i_{S_{1_{pk}}} = \frac{V_{in} \cdot DT_s}{L_{m_1}} \tag{2}$$

where V_{in} is the input voltage, D is the duty cycle, T_s is the switching period, and L_{m_1} is the magnetizing inductance of transformer $N_{12}-N_{34}$. Consequently, the flux densities B'_{o_1} and B'_{o_2} also reach their peak values $B'_{o_{pk}}$, which are given by (3):

$$B'_{o_{pk}} = \frac{L_{m_1} i_{S_{1_{pk}}}}{N_{12} A_o} \tag{3}$$

with N_{12} as the number of turns in the primary side of the active transformer and A_o is the cross-sectional area of the outer leg, where the transformer $N_{12} - N_{34}$ is wound.

3) Mode 3 ($t_3 < t < t_4$, Fig. 3(c)): Mode 3 starts right after transformer $N_{12} - N_{34}$ finishes charging. At t_3, Switch S_1 turns off, initiating the discharge phase of transformer $N_{12} - N_{34}$. The magnetizing current $i_{L_{m_1}}$, diode current i_{D_1}, and flux densities B'_{o_1} and B'_{o_2} begin to decrease linearly. Transformer $N_{56} - N_{78}$ remains inactive in this period.

In Modes 1-3, the magnetomotive force and magnetic resistance of both outer legs are equal. The magnetic flux in the center leg is canceled. This causes no electromotive force generated in transformer $N_{56}-N_{78}$. Therefore, the transformer formed by the outer legs operates independently from the transformer in the center leg. The integrated transformer is equivalent to that of Fig. 5(b) and its equivalent magnetic circuit is shown in Fig. 6(a).

4) Mode 4 ($t_4 < t < t_5$, Fig. 3(d)): Mode 4 operates similarly to the positive cycle. Switch S_2 turns on at t_4 and S_1 remains off. Transformer $N_{56}-N_{78}$ stores energy in its magnetizing inductance. Its current $i_{L_{m_2}}$ and the corresponding flux densities B''_{o_1}, B''_c, and B''_{o_2} linearly increase. Simultaneously, transformer $N_{12} - N_{34}$ completes its discharge phase, with $i_{L_{m_1}}$ and i_{D_1} decreasing

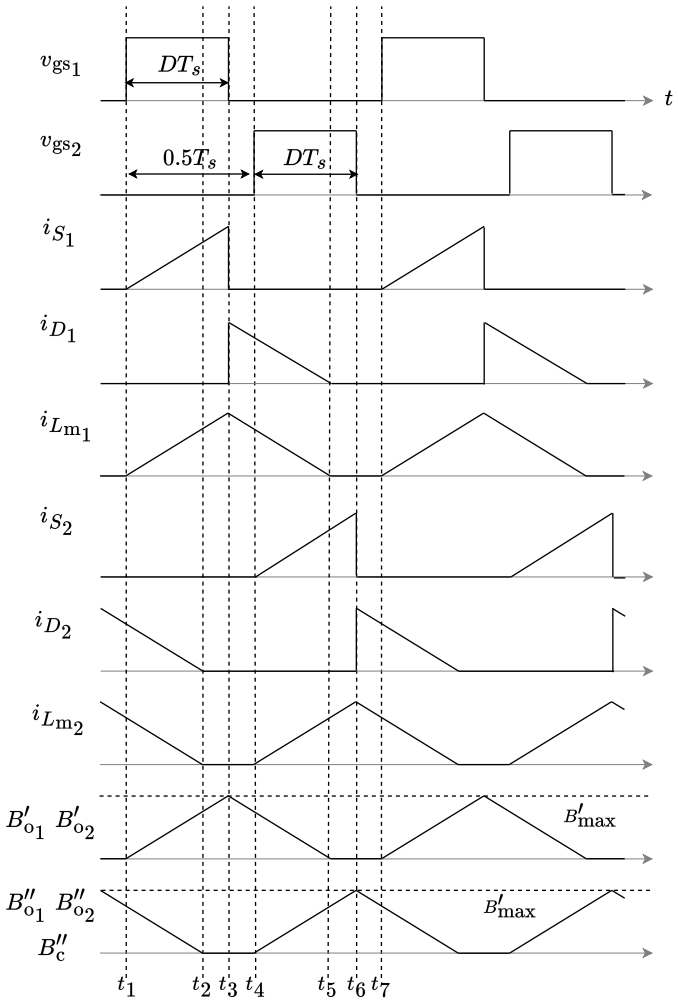


Fig. 4. Key waveforms in the operation of the two-phase interleaved flyback rectifier with the proposed transformer, with $D \leq 0.5\frac{nV_o}{V_{in_{max}}}$.

to zero at t_5.

5) Mode 5 ($t_5 < t < t_6$, Fig. 3(e)): Mode 5 starts after the discharge of transformer $N_{12}-N_{34}$ completes. Now, with Switch S_2 still on, transformer $N_{56}-N_{78}$ continues storing energy until $i_{L_{m_2}}$ reaches its peak value at t_6. The peak primary current in the negative cycle $i_{S_{2_{pk}}}$ is given by (4):

$$i_{S_{2_{pk}}} = \frac{V_{in} \cdot DT_s}{L_{m_2}} \tag{4}$$

where L_{m_2} is the magnetizing inductance of transformer $N_{56} - N_{78}$. Consequently, the flux densities B''_{o_1}, B''_c, and B''_{o_2} also reach their peak values. The peak value of $B''_{c_{pk}}$ is given by (5):

$$B''_{c_{pk}} = B''_{o_{pk}} = \frac{L_{m_2} i_{S_{2_{pk}}}}{N_{56} A_c} \tag{5}$$

with N_{56} as the number of turns in the primary side of the active transformer and A_c is the cross-sectional area of the center leg where the transformer $N_{56} - N_{78}$ is wound.

6) Mode 6 ($t_6 < t < t_7$, Fig. 3(f)): Switch S_2 turns off, initiating the discharge phase of transformer $N_{56} - N_{78}$. The currents $i_{L_{m_2}}, i_{D_2}$, and flux densities B''_{o_1}, B''_c, and B''_{o_2} begin to decrease linearly. Transformer $N_{12} - N_{34}$ remains inactive.

For Modes 4-6, since the magnetic resistance of both outer legs is equal, the magnetic flux is divided equally into both outer legs. Furthermore, since the number of turns of both outer legs is equal and wound in an opposite polarity, the electromotive force generated in $N_{12} - N_{34}$ is canceled. Therefore, the transformer formed by the center leg operates independently from the transformer in the outer leg, and the integrated transformer is equivalent to that of Fig. 5(c). The equivalent magnetic circuit for this case is shown in Fig. 6(b).

The proposed winding structure achieves zero magnetic coupling between the two transformers. Consequently, the distinct magnetic circuits share a single core without magnetic

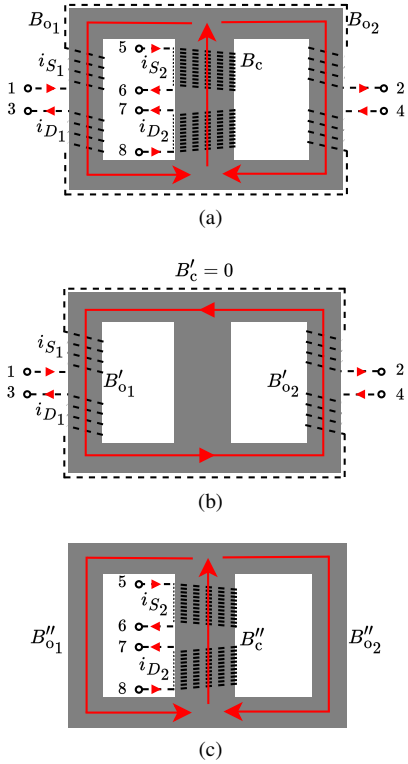


Fig. 5. The proposed integrated transformer during the two phases of the converter, with the flux density components in each leg illustrated in (a). (b) is realized during the positive AC cycle, while (c) is realized during the negative cycle.

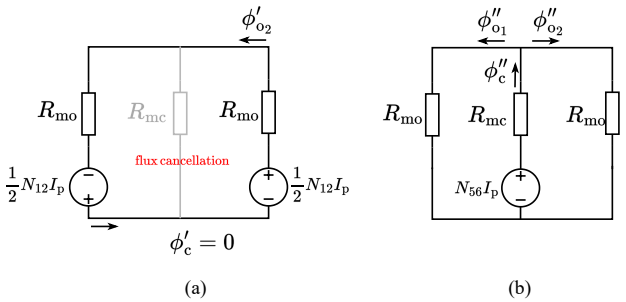


Fig. 6. Equivalent magnetic circuits for the proposed transformer. (a) shows the equivalent magnetic circuit for $N_{12}-N_{34}$, while (b) shows the equivalent magnetic circuit for $N_{56} - N_{78}$.

interference. Each transformer maintains independent operation despite the physical integration. This structure eliminates the need for two discrete magnetic components. As a result, the total volume of the magnetic components is significantly reduced.

B. Saturation Magnetic Flux Density Analysis

The circuit operation is classified into three patterns determined by D and the boundary condition $D \leq 0.5\frac{nV_\mathrm{o}}{V_{\mathrm{in}_\mathrm{max}}}$. This classification, illustrated in Fig. 7, arises because the integrated single-core structure permits simultaneous magnetizing current flow in multiple active windings, resulting in distinct flux superposition compared to conventional discrete transformers. The manner of superposition due to the magnetizing current can be controlled by t_dis. Consequently, the saturation magnetic flux density B_max is a critical parameter for the transformer geometry. It can be expressed in terms of the saturation magnetic flux density in each phase, denoted as B'_max and B''_max, with $B'_\mathrm{max} = B''_\mathrm{max}$ since $L_{\mathrm{m}_1} = L_{\mathrm{m}_2}$. The flux density in each leg is examined further using the principle of superposition, with the expressions of the superimposed flux densities in the left, right and center legs shown in (6)-(8), represented as B_{o_1}, B_{o_2} and B_c respectively:

$$B_{\mathrm{o}_1} = B'_{\mathrm{o}_1} + B''_{\mathrm{o}_1} \tag{6}$$

$$B_{\mathrm{o}_2} = B'_{\mathrm{o}_2} - B''_{\mathrm{o}_2} \tag{7}$$

$$B_\mathrm{c} = B''_\mathrm{c} \tag{8}$$

Case 1 (Fig. 7(a)), $D \leq 0.5\frac{nV_\mathrm{o}}{V_{\mathrm{in}_\mathrm{max}}}$*:* The waveforms in Fig. 4 represent the circuit operation for Case 1. As shown, flux cancellation does not occur in the outer legs; however, the magnitude of the peak flux density remains uniform across all legs. If the magnetizing current of one phase reaches zero before the magnetizing current of the other phase reaches its peak, then $B_\mathrm{max} = B'_\mathrm{max}$. This condition is satisfied when D is expressed by (9).

$$D \leq 0.5\frac{nV_\mathrm{o}}{V_{\mathrm{in}_\mathrm{max}}} \tag{9}$$

Case 2 (Fig. 7(b)), $0.5\frac{nV_\mathrm{o}}{V_{\mathrm{in}_\mathrm{max}}} < D \leq 0.5$*:* In this case, the magnetizing current of one phase does not reach zero before the subsequent phase reaches its peak. The saturation flux density across all phases can be derived as follows. At t_2, $B'_{\mathrm{o}_1} = B'_\mathrm{max}$. Moreover, $B''_{\mathrm{o}_1}(t_2) = B'_{\mathrm{o}_1}(t_5)$ and is expressed in (10):

$$B''_{\mathrm{o}_1}(t_2) = B'_{\mathrm{o}_1}(t_5) = B'_\mathrm{max} - \frac{B'_\mathrm{max}}{t_6 - t_2}(t_5 - t_2) \tag{10}$$

with $t_5 = 0.5T_\mathrm{s} + t_2$ and $t_2 = DT_\mathrm{s}$. The peak $i_{L_{\mathrm{m}_{1\mathrm{pk}}}}$ can be rewritten as in (11):

$$i_{L_{\mathrm{m}_{1\mathrm{pk}}}} = \frac{V_{\mathrm{in}_\mathrm{max}}}{L_\mathrm{m}}DT_\mathrm{s} = \frac{nV_\mathrm{o}}{L_\mathrm{m}}(t_6 - t_2) \tag{11}$$

In this case, (9) is not satisfied, and this results in B_max in the left outer leg exceeding B'_max in the center leg, as expressed in (12):

$$B_\mathrm{max} = \left(2 - \frac{0.5nV_\mathrm{o}}{V_{\mathrm{in}_\mathrm{max}}D}\right)B'_\mathrm{max} \tag{12}$$

Case 3 (Fig. 7(c)), $0.5 < D$*:* In this case, B_max in the left outer leg consistently exceeds B'_max. By analysis similar to Case 2, B_max is derived as (13):

$$B_\mathrm{max} = \left(1 + \frac{D - 0.5}{D}\right)B'_\mathrm{max} \tag{13}$$

Because the flyback is interleaved, it is expected that the rectifier will primarily operate at $D \leq 0.5$, which operates in the conditions set by Case 1 or 2.

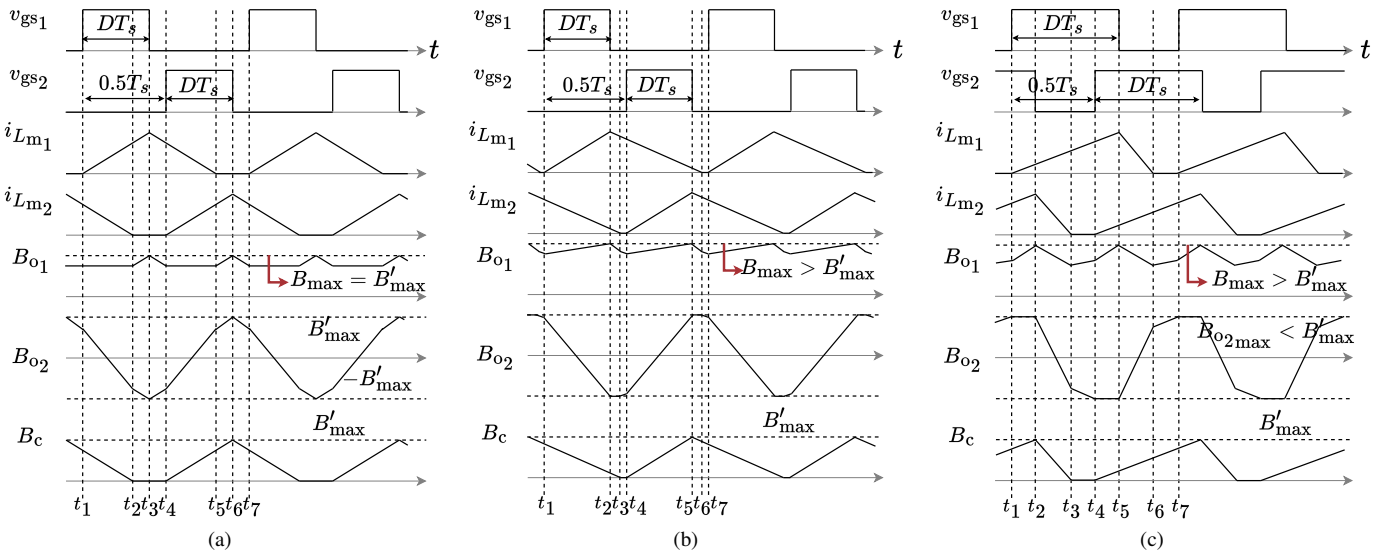


Fig. 7. Theoretical, superimposed magnetic flux density waveforms in the integrated transformer during circuit operations: (a) Case 1, (b) Case 2, and (c) Case 3.

C. Design Method of the Proposed Integrated Transformer

A turns ratio n is selected based on the maximum voltage ratings of the switches. The maximum operating duty cycle D is selected in consideration of the desired operation discussed in Section II-B. The magnetizing inductance, $L_{\mathrm{m}_1} = L_{\mathrm{m}_2} = L_{\mathrm{m}}$ is calculated from the circuit specifications. The saturation magnetic flux density per phase $B'_{\max}$ is calculated from the $B_{\max}$ given by the chosen core material, considering the maximum D. A core is chosen using the AP method.

The number of turns is calculated using $B'_{\max}$ and the peak primary side current $I_{\mathrm{pri}_{\mathrm{pk}}}$ as shown in (14) and (15):

$$\frac{N_{12}}{n} = N_{34} \geq \frac{L_{\mathrm{m}} I_{\mathrm{pri}_{\mathrm{pk}}}}{n A_{\mathrm{mo}} B'_{\max}} \tag{14}$$

$$\frac{N_{56}}{n} = N_{78} \geq \frac{L_{\mathrm{m}} I_{\mathrm{pri}_{\mathrm{pk}}}}{n A_{\mathrm{mc}} B'_{\max}} \tag{15}$$

The magnetic resistances of the outer legs R_{mo} and the center leg R_{mc} are obtained from a magnetic circuit analysis of the integrated transformer in its separate phases. The transformer $N_{12} - N_{34}$ is only made of the outer legs as shown in Fig. 8(a), while $N_{56} - N_{78}$ is made of all the legs. The magnetizing inductance L_{m} in terms of the number of turns and magnetic resistances for $N_{12} - N_{34}$ and $N_{56} - N_{78}$ are shown in (16) and (17) respectively.

$$L_{\mathrm{m}} = \frac{N_{12}^2}{2R_{\mathrm{mo}}} \tag{16}$$

$$L_{\mathrm{m}} = \frac{N_{56}^2}{R_{\mathrm{mc}} + 0.5R_{\mathrm{mo}}} \tag{17}$$

The magnetic resistances R_{mo} and R_{mc} are obtained from (18) and (19):

$$R_{\mathrm{mo}} = \frac{N_{12}^2}{2L_{\mathrm{m}}} \tag{18}$$

$$R_{\mathrm{mc}} = \frac{(2N_{56} - N_{12})(2N_{56} + N_{12})}{4L_{\mathrm{m}}} \tag{19}$$

Finally, an air gap was also inserted in each of the legs to provide the desired resistance. The magnetic path lengths, l_{mo} and l_{mc}, and cross-sectional areas, A_{mo} and A_{mc}, of the outer and center legs respectively are shown in Fig. 8(b). Non-equal length air gaps were placed in the transformer. There are two cases, one where the air gaps in the outer legs $l_{\mathrm{mo}_{\mathrm{air}}}$ are longer than the air gap in the center leg $l_{\mathrm{mc}_{\mathrm{air}}}$, and one where $l_{\mathrm{mc}_{\mathrm{air}}}$ is longer than $l_{\mathrm{mo}_{\mathrm{air}}}$.

Case 1: $l_{\mathrm{mo}_{\mathrm{air}}} > l_{\mathrm{mc}_{\mathrm{air}}}$: Given the setup in Fig. 9(a), R_{mo} and R_{mc} are shown in (20) and (21) respectively with $l_{\mathrm{mo}_{\mathrm{air}}}$ and $l_{\mathrm{mc}_{\mathrm{air}}}$ in (22) and (23) respectively.

$$R_{\mathrm{mo}} = \frac{l_{\mathrm{mo}} - l_{\mathrm{mo}_{\mathrm{air}}} + l_{\mathrm{mc}_{\mathrm{air}}}}{\mu_r \mu_0 A_{\mathrm{mo}}} + \frac{l_{\mathrm{mo}_{\mathrm{air}}}}{\mu_0 A_{\mathrm{mo}}} \tag{20}$$

$$R_{\mathrm{mc}} = \frac{l_{\mathrm{mc}}}{\mu_r \mu_0 A_{\mathrm{mc}}} + \frac{l_{\mathrm{mc}_{\mathrm{air}}}}{\mu_0 A_{\mathrm{mc}}} \tag{21}$$

$$l_{\mathrm{mo}_{\mathrm{air}}} = \frac{\mu_0(\mu_r A_{\mathrm{mo}} R_{\mathrm{mo}} - A_{\mathrm{mc}} R_{\mathrm{mc}}) + \frac{l_{\mathrm{mc}}}{\mu_r} - l_{\mathrm{mo}}}{\mu_r - 1} \tag{22}$$

$$l_{\mathrm{mc}_{\mathrm{air}}} = \mu_0 A_{\mathrm{mc}} R_{\mathrm{mc}} - \frac{l_{\mathrm{mc}}}{\mu_r} \tag{23}$$

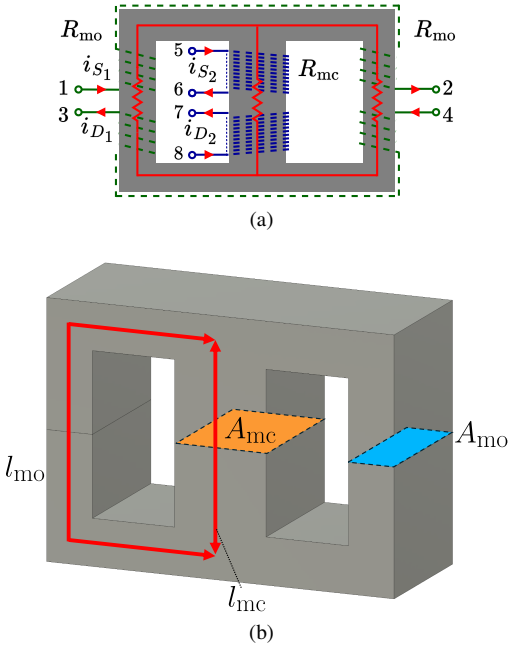


Fig. 8. EE core configuration: (a) magnetic resistances of the transformer (R_{mo} for the outer legs, R_{mc} for the center leg), and (b) effective cross-sectional area (A_{mo} for the outer legs, A_{mc} for the center leg) and effective magnetic path length l_{mo} for the outer legs, l_{mc} for the center leg.

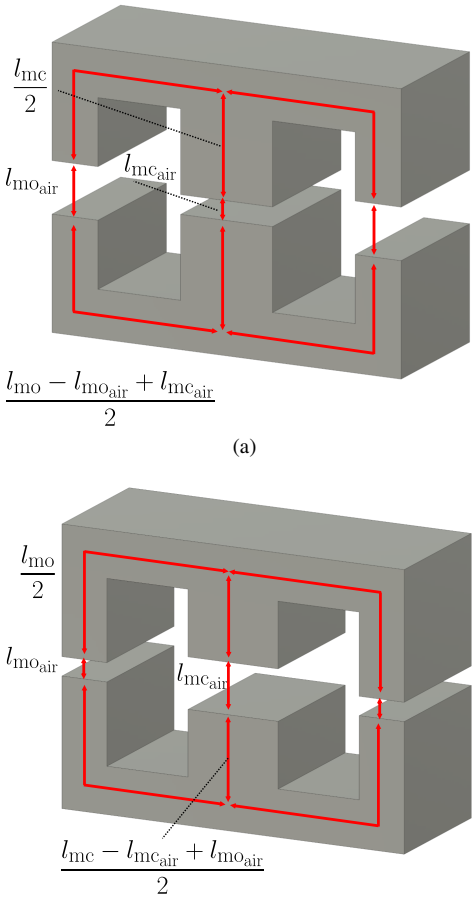


Fig. 9. EE core configuration with air gaps: (a) setup when $l_{\mathrm{mo}_{\mathrm{air}}} > l_{\mathrm{mc}_{\mathrm{air}}}$ and (b) setup when $l_{\mathrm{mo}_{\mathrm{air}}} < l_{\mathrm{mc}_{\mathrm{air}}}$.

Case 2: $l_{\mathrm{mo_{air}}} < l_{\mathrm{mc_{air}}}$: Given the setup in Fig. 9(b), R_{mo} and R_{mc} are shown in (24) and (25) respectively, $l_{\mathrm{mo_{air}}}$ and $l_{\mathrm{mc_{air}}}$ in (26) and (27) respectively.

$$R_{\mathrm{mo}} = \frac{l_{\mathrm{mo}}}{\mu_r \mu_0 A_{\mathrm{mo}}} + \frac{l_{\mathrm{mo_{air}}}}{\mu_0 A_{\mathrm{mo}}} \tag{24}$$

$$R_{\mathrm{mc}} = \frac{l_{\mathrm{mc}} - l_{\mathrm{mc_{air}}} + l_{\mathrm{mo_{air}}}}{\mu_r \mu_0 A_{\mathrm{mc}}} + \frac{l_{\mathrm{mc_{air}}}}{\mu_0 A_{\mathrm{mc}}} \tag{25}$$

$$l_{\mathrm{mo_{air}}} = \mu_0 A_{\mathrm{mo}} R_{\mathrm{mo}} - \frac{l_{\mathrm{mo}}}{\mu_r} \tag{26}$$

$$l_{\mathrm{mc_{air}}} = \frac{\mu_0(\mu_r A_{\mathrm{mc}} R_{\mathrm{mc}} - A_{\mathrm{mo}} R_{\mathrm{mo}}) + \frac{l_{\mathrm{mo}}}{\mu_r} - l_{\mathrm{mc}}}{\mu_r - 1} \tag{27}$$

III. Simulation Verification of Proposed Design

A. Simulation Verification of the Proposed Design Method

To verify the operation of the integrated transformer, a simulation of the two-phase interleaved flyback rectifier was first performed in PLECS Standalone 4.9.4, with the diagram shown in Fig. 10(a). Clamped RCD Snubbers ($C_{\mathrm{sn_1}}, D_{\mathrm{sn_1}}, R_{\mathrm{sn_1}}$ and $C_{\mathrm{sn_2}}, D_{\mathrm{sn_2}}, R_{\mathrm{sn_2}}$) were also added for protection against surge voltage [26], [27] caused by the leakage inductance and the switch capacitance at the turn-off of S_1 and S_2. An integrated transformer, shown in Fig. 10(b), was also designed using the proposed method. The circuit and transformer were designed with the condition given by (9), and will thus be verified in this study. The specifications of the two-phase interleaved flyback rectifier and transformer for the simulation verification are shown in Tables I and II respectively. The duty cycle is calculated using (28) [6].

$$D = \frac{2}{V_{\mathrm{in_{pk}}}} \sqrt{\frac{0.5 L_{\mathrm{m}} P_{\mathrm{o}}}{T_{\mathrm{s}}}} \tag{28}$$

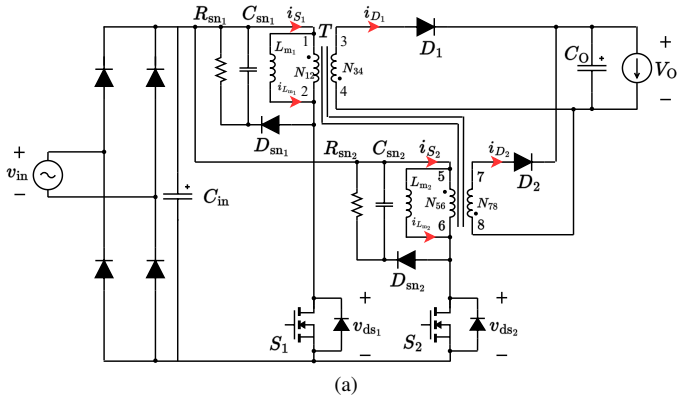


Fig. 10. Prototype two-phase interleaved flyback rectifier (a) with the proposed integrated transformer (b).

TABLE I
Simulation Specifications of the Prototype Two-Phase Interleaved Flyback Rectifier

Item	Symbol	Value
AC input voltage	v_{in}	100 V V_{rms}
Maximum rated power	P_{O}	240 W
Output voltage	V_{O}	48 V
Switching frequency	f_{s}	100 kHz
Input capacitor	C_{in}	1 µF
Output capacitor	C_{O}	2 x 1mF (parallel)
Snubber capacitor	$C_{\mathrm{sn_1}}, C_{\mathrm{sn_2}}$	2.2 nF
Snubber resistor	$R_{\mathrm{sn_1}}, R_{\mathrm{sn_2}}$	30 kΩ

TABLE II
Simulation Specifications of the Prototype Integrated Transformer

Item	Symbol	Value
Magnetic core	-	OP44721EC
Relative permeability	μ_r	2500
Magnetic path length (center leg)	l_{mc}	31.7 mm
Magnetic path length (outer leg)	l_{mo}	71.4 mm
Cross-sectional area (center leg)	A_{mc}	243.4 mm^2
Cross-sectional area (outer leg)	A_{mo}	117.6 mm^2
Magnetizing inductance	$L_{\mathrm{m_1}}, L_{\mathrm{m_2}}$	82.8 µH
Turn ratio	n	3
Number of turns	$N_{12} : N_{34}$	24:8
	$N_{56} : N_{78}$	18:6
Center leg air gap	$l_{\mathrm{mc_{air}}}$	0.650 mm
Outer leg air gap	$l_{\mathrm{mo_{air}}}$	0.500 mm
Coupling coefficient of N_{12}, N_{34}	$k_{N_{12}-N_{34}}$	0.990
Coupling coefficient of N_{56}, N_{78}	$k_{N_{56}-N_{78}}$	0.990
Coupling coefficient of N_{12}, N_{56}	$k_{N_{12}-N_{56}}$	0.000

B. Simulation Results

Fig. 11 shows waveforms from the simulation testing of the two-phase interleaved flyback rectifier with the proposed transformer at 240 W. Key waveforms of the circuit were observed from the simulation, which include $v_{\mathrm{gs_1}}$ and $v_{\mathrm{gs_2}}$, $i_{L_{\mathrm{m_1}}}$ and $i_{L_{\mathrm{m_2}}}$, and the fluxes $B_{\mathrm{o_1}}$, $B_{\mathrm{o_2}}$, and B_{c}. Similar waveforms to that of Fig. 4 have been observed, confirming that the system operates exactly as expected from the theoretical analysis. Flux cancellation is observed in the center leg during the time that $N_{56} - N_{78}$ is inactive, as denoted by the interval when $i_{L_{\mathrm{m_2}}} = 0$. The magnetizing currents $i_{L_{\mathrm{m_1}}}$ and $i_{L_{\mathrm{m_2}}}$ subside to zero prior to the subsequent phase reaching its peak. Moreover, the increase and decrease of fluxes were all linear, therefore confirming that there is indeed no magnetic coupling interference. This confirms the validity of the theoretical operation proposed in Section II.

IV. Size Reduction Effect of the Proposed Transformer

This section examines the size reduction effect by the integrated transformer. For this study, the conventional transformer is assumed to be two separate transformers that facilitate isolation between the primary and secondary side. A representation is shown in Fig. 12(a), and its windings are represented in Fig. 13(a). The proposed integrated transformer is represented in Fig. 12(b) and its windings are represented in Fig. 13(b). The volume is compared by using the AP method. The AP method utilizes the product between the core cross-sectional area, A_{c}, and the window-area, A_{w}, and is a fundamental core size for

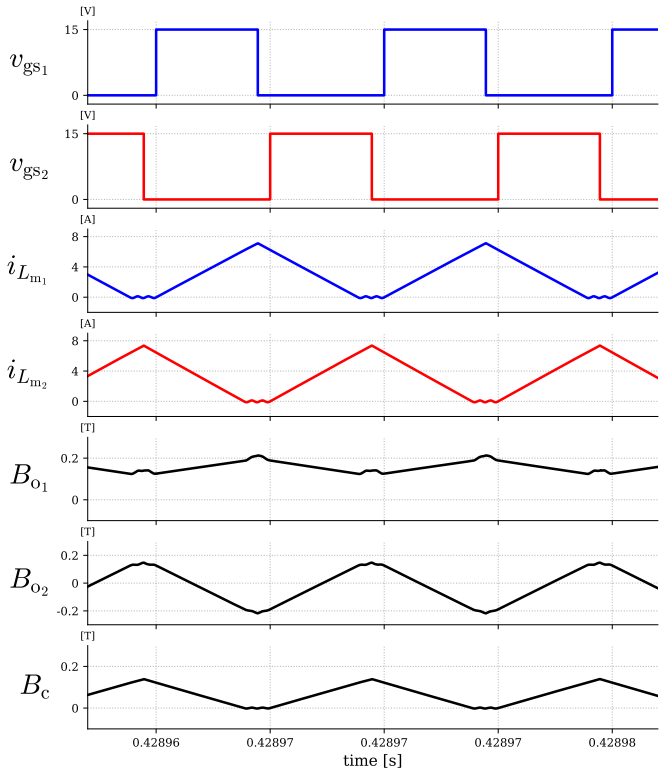


Fig. 11. Key simulation waveforms of the proposed two-phase interleaved flyback rectifier at 240 W.

the application [28]. It can be expressed both in terms of the transformer geometric parameters and the circuit parameters, as will be demonstrated in the following analysis.

The EE-core was redesigned in such a way such that A_w and A_c are squares to allow for simpler analysis, as shown in Fig. 14 [29]. Here, A_c is twice the area of the outer leg.

For the conventional transformer, A_c is expressed in (29). Here, N_p is the number of turns wound on the center leg of

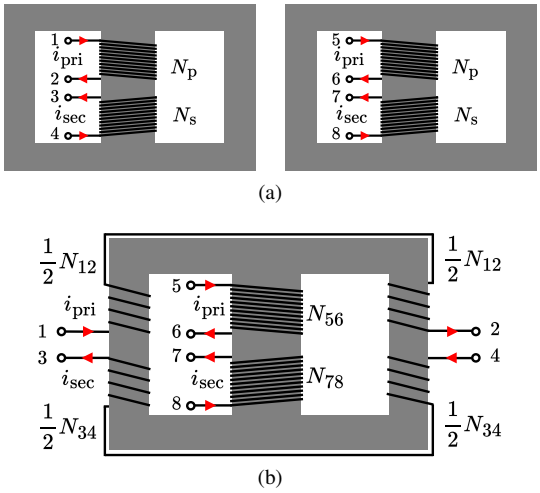


Fig. 12. Two separate transformers (a) and proposed integrated transformer (b) used in the volume analysis.

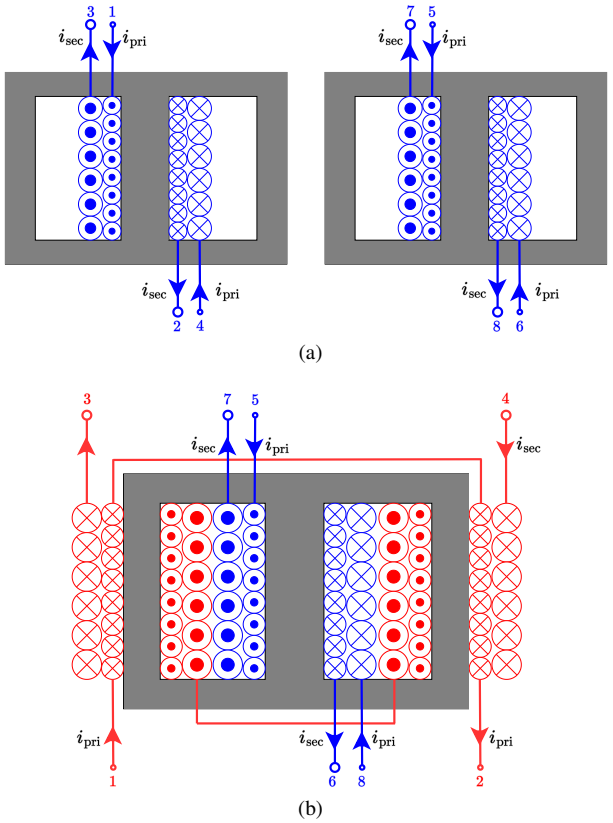


Fig. 13. Two separate transformers (a) and the proposed integrated transformer (b) used in the volume analysis, with the windings represented as well.

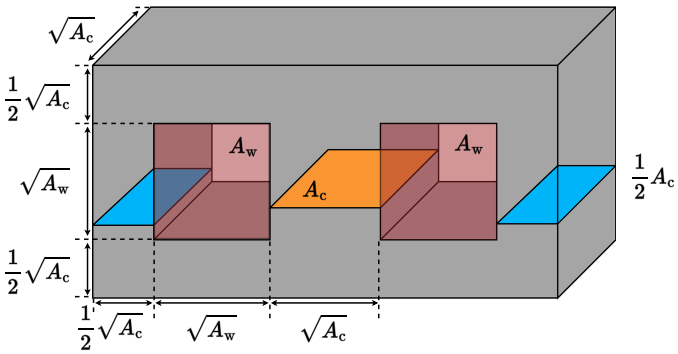


Fig. 14. Simplified representation of one EE-core used in the volume analysis.

the conventional transformer.

$$A_{c_{conv}} = \frac{L_m I_{pri_{pk}}}{B_{max} N_p} \tag{29}$$

For the integrated transformer A_c is expressed in (30).

$$A_{c_{integ}} = \frac{L_m I_{pri_{pk}}}{B_{max} N_{56}} \tag{30}$$

The window area A_w can be expressed in terms of winding area and the occupancy ratio K_u, as seen in (31) for the conventional transformer:

$$A_{w_{conv}} K_u = N_p \frac{I_{pri_{rms}}}{J_m} + N_s \frac{I_{sec_{rms}}}{J_m} \tag{31}$$

which simplifies to:

$$A_{\mathrm{w_{conv}}} = \frac{N_{\mathrm{p}}}{J_{\mathrm{m}} K_{\mathrm{u}}} \left(I_{\mathrm{pri_{rms}}} + \frac{I_{\mathrm{sec_{rms}}}}{n} \right) \tag{32}$$

For the integrated transformer, taking into account that the core window has half the total turns of the transformer $N_{12} - N_{34}$, A_{w} can be expressed as seen in (33).

$$A_{\mathrm{w_{integ}}} K_{\mathrm{u}} = \left(N_{56} \frac{I_{\mathrm{pri_{rms}}}}{J_{\mathrm{m}}} + N_{78} \frac{I_{\mathrm{sec_{rms}}}}{J_{\mathrm{m}}} \right) + \frac{1}{2} \left(N_{12} \frac{I_{\mathrm{pri_{rms}}}}{J_{\mathrm{m}}} + N_{34} \frac{I_{\mathrm{sec_{rms}}}}{J_{\mathrm{m}}} \right) \tag{33}$$

The total volume of the EE core in Fig. 14 expressed in (34), where x is a constant that denotes $A_{\mathrm{c}}/A_{\mathrm{w}}$. The volume corresponding to the conventional and integrated transformer is obtained using their respective A_{c} and A_{w}.

$$\mathrm{Vol} = 2AP^{\frac{3}{4}}(2x^{\frac{1}{4}} + x^{\frac{3}{4}} + x^{-\frac{1}{4}}) \tag{34}$$

The rms expressions of the primary and secondary side current of the interleaved flyback rectifier are given in (35) and (36) [6].

$$I_{\mathrm{pri_{rms}}} = \frac{V_{\mathrm{in_{max}}}}{L_{\mathrm{m}} f_{\mathrm{s}}} \sqrt{\frac{D^3}{6}} \tag{35}$$

$$I_{\mathrm{sec_{rms}}} = \frac{V_{\mathrm{in_{max}}}}{L_{\mathrm{m}} f_{\mathrm{s}}} \sqrt{\frac{4nD^3}{9\pi} \cdot \frac{V_{\mathrm{in_{max}}}}{V_{\mathrm{o}}}} \tag{36}$$

The proposed design is estimated to reduce core volume by 56% under the assumptions of the AP-based analysis using the designated circuit specfications, with the comparison shown in Table III. This assumes that the integrated core geometry is custom-tailored to the specific AP requirements derived from the circuit specifications, rather than being constrained by standard commercial core shapes. While the experimental prototype described in Section V utilizes a commercial core (Magnetics 0P44721EC) for proof-of-concept, this theoretical analysis confirms that a purpose-built integrated core would offer significant improvements in power density compared to the conventional discrete implementation.

TABLE III
VOLUME ANALYSIS OF THE CONVENTIONAL AND PROPOSED TRANSFORMER

Item	Symbol	Conventional	Proposed
Maximum input voltage	$V_{\mathrm{in_{max}}}$	240 V V_{rms}	
Output voltage	V_{O}	48 V	
Maximum duty cycle	D	0.47	
Switching frequency	f_{s}	100 kHz	
Magnetizing inductance	L_{m}	82.8 µH	
Turn ratio	n	3	
Number of turns (Conv.)	$N_{\mathrm{p}} : N_{\mathrm{s}}$	24:8	–
Number of turns (Prop.)	$N_{12} : N_{34}$	–	24:8
	$N_{56} : N_{78}$	–	18:6
Primary side peak current	$I_{\mathrm{pri,pk}}$	8.02 A	
Primary side rms current	$I_{\mathrm{pri,rms}}$	5.39 A	
Secondary side rms current	$I_{\mathrm{sec,rms}}$	12.53 A	
Saturation flux density	B_{max}	0.3 T	
Maximum current density	J_{m}	5 A/mm^2	
Occupancy ratio	K_{u}	0.3	
Area Product	AP	14128.75 mm^4	23547.91 mm^4
Core Volume	Vol	21106.38 mm^3	9398.46 mm^3

V. EXPERIMENTAL VERIFICATION OF PROPOSED DESIGN

A. Prototype Evaluation Circuit and Prototype Transformer

A prototype two-phase interleaved flyback rectifier, shown in Fig. 10(a) operating in DCM was built to verify the operation of the integrated transformer. The prototype integrated transformer, shown in Fig. 10(b), was also designed using the proposed method. The specifications of the two-phase interleaved flyback rectifier and transformer are shown in Tables IV and V respectively. The transformer parameters were characterized using a Hioki IM3536 LCR meter at 100 kHz. The coupling coefficients of the transformers $k_{N_{12}-N_{34}}$, $k_{N_{56}-N_{78}}$, and $k_{N_{12}-N_{56}}$ were derived from the mutual inductance obtained after connecting the respective transformer windings in series-aiding and series-opposing configurations. The coupling coefficient between the two flyback transformers $k_{N_{12}-N_{56}}$ is 0.0005, showing that the coupling was almost eliminated.

B. Experimental Results

Fig. 15(a) shows the 100 W experimental waveforms at 4 ms/div from the two-phase interleaved flyback rectifier with the proposed transformer, demonstrating power factor correction. Fig. 15(b) presents 2 µs/div waveforms. No current is induced in either switch except during the resonance between the leakage inductance and the parasitic capacitance of the switch. During the inactive period of the transformer, secondary diode side current is zero. Moreover, after $i_{\mathrm{S_1}}$ and $i_{\mathrm{S_2}}$ reached zero when S_1 and S_2 switched off, the currents

TABLE IV
SPECIFICATIONS OF THE PROTOTYPE TWO-PHASE INTERLEAVED FLYBACK RECTIFIER

Item	Symbol	Value / Component
AC input voltage	v_{in}	100-240 V V_{rms}
Maximum rated power	P_{O}	240 W
Output voltage	V_{O}	48 V
Switching frequency	f_{s}	100 kHz
Bridge diode	BD	CD-DF408S
Input capacitor	C_{in}	1 µF
Switch	S_1, S_2	STF15N80k5
Secondary diode	D_1, D_2	STPSC30H12CWL
Output capacitor	C_{O}	2 x 1mF, in parallel
Snubber capacitor	$C_{\mathrm{sn_1}}, C_{\mathrm{sn_2}}$	2.2 nF
Snubber resistor	$R_{\mathrm{sn_1}}, R_{\mathrm{sn_2}}$	30 kΩ
Snubber diode	$D_{\mathrm{sn_1}}, D_{\mathrm{sn_2}}$	VS-E7MH0112-M3/1

TABLE V
SPECIFICATIONS OF THE PROTOTYPE INTEGRATED TRANSFORMER

Item	Symbol	Value
Magnetizing inductance (Target)	L_{m}	82.8 µH
Magnetizing inductance of $N_{12} - N_{34}$	$L_{\mathrm{m_1}}$	86.1 µH
Magnetizing inductance of $N_{56} - N_{78}$	$L_{\mathrm{m_2}}$	75.1 µH
Turn ratio	n	3
Magnetic core	-	OP44721EC
Number of turns	$N_{12} : N_{34}$	24:8
	$N_{56} : N_{78}$	15:5
Center leg air gap	$l_{\mathrm{mc_{air}}}$	0.650 mm
Outer leg air gap	$l_{\mathrm{mo_{air}}}$	0.500 mm
Coupling coefficient of N_{12}, N_{34}	$k_{N_{12}-N_{34}}$	0.970
Coupling coefficient of N_{56}, N_{78}	$k_{N_{56}-N_{78}}$	0.979
Coupling coefficient of N_{12}, N_{56}	$k_{N_{12}-N_{56}}$	0.0005

remained at zero. Therefore, the two flyback converters operate independently without magnetic coupling interference. The same phenomenon was observed at 240 W two-phase drive, as shown in Fig. 15(c). Single-phase drive at lower power at 32.5 W is also feasible as shown in Fig. 15(d).

VI. Conclusion

In this paper, an integrated transformer was successfully designed for a two-phase interleaved flyback rectifier for USB PD applications. The proposed winding structure for the integrated transformer prevents coupling between the two flyback transformers and achieves simple manufacturability through the use of a commercial EE-core. A coupling coefficient of 0.0005 was achieved with the prototype. Independent flyback circuit operation without magnetic interference due to coupling was verified through simulation and experiment up to 240 W. Through the proposed design, two discrete transformers are integrated into a single magnetic component. The structure is estimated to enable up to a 56% downsizing of magnetic components compared to conventional discrete transformers.

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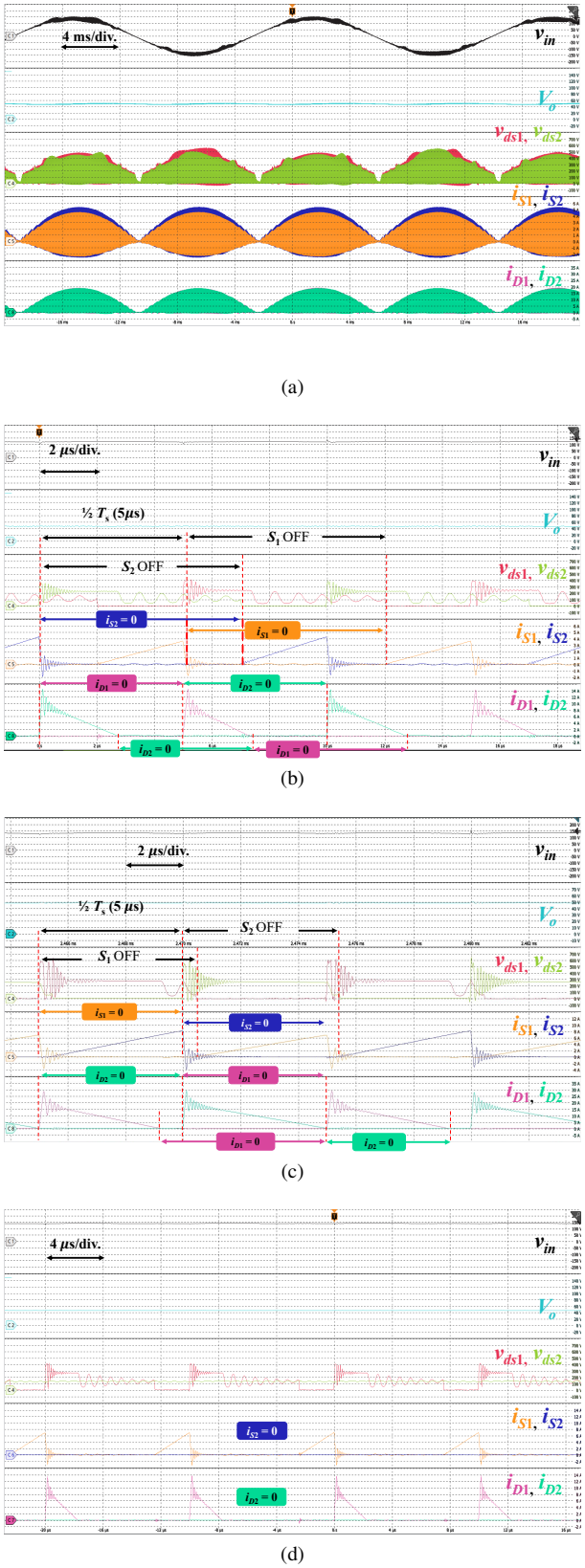


Fig. 15. Experimental waveforms of a two-phase interleaved flyback rectifier with the proposed integrated transformer at (a) & (b): 100 W, two-phase AC drive, (c): 240 W, two-phase AC drive, (d): 32.5 W, single-phase AC drive. For (a) and (b): v_{in}: 50 V/div., V_{o}: 10 V/div., $v_{\mathrm{ds}_1}, v_{\mathrm{ds}_2}$: 100 V/div., $i_{\mathrm{S}_1}, i_{\mathrm{S}_2}$: 1 A/div., $i_{\mathrm{D}_1}, i_{\mathrm{D}_2}$: 5 A/div.; For (c): v_{in}: 50 V/div., V_{o}: 10 V/div., $v_{\mathrm{ds1}}, v_{\mathrm{ds2}}$: 100 V/div., $i_{\mathrm{S}_1}, i_{\mathrm{S}_2}$: 2 A/div., $i_{\mathrm{D}_1}, i_{\mathrm{D}_2}$: 5 A/div.; For (d): v_{in}: 50 V/div., V_{o}: 20 V/div., $v_{\mathrm{ds}_1}, v_{\mathrm{ds}_2}$: 100 V/div., $i_{\mathrm{S}_1}, i_{\mathrm{S}_2}$: 2 A/div., $i_{\mathrm{D}_1}, i_{\mathrm{D}_2}$: 2 A/div.

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